Water Footprint of Artificial Intelligence and its Repercussions on Environmental Health: A Critical Analysis within the One Health Framework

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***Abstract*—The rise of large artificial intelligence models, particularly language and diffusion models, is rapidly increasing computational demand in data centers. While their carbon emissions are increasingly discussed, their water footprint remains less visible. Training and inference consume freshwater directly, through evaporative cooling, and indirectly, via electricity generation. This consumption can aggravate local water stress, reduce water available for households, agriculture, and ecosystems, and consequently amplify human, animal, and environmental health risks. Through the One Health framework, this article analyzes the mechanisms linking AI to planetary health, with special attention to geographical inequalities when data centers are located in water-constrained regions. We advocate for mandatory water audits, transparency standards, and watershed-sensitive AI governance.**

***Index Terms*—water footprint, artificial intelligence, One Health, data centers, water stress, environmental equity, planetary health**

I. Introduction

Generative Artificial Intelligence (AI) has become a critical infrastructure for research, industry, and public services. Large language models such as GPT-4, Llama, and Gemini, as well as diffusion models used to generate images, videos, or molecular structures, rely on clusters of graphics processing units and specialized accelerators running for weeks or months [1]–[3]. The literature on the environmental impact of AI initially focused on electricity and carbon, as training a large model can mobilize thousands of megawatt-hours and depends heavily on the regional electricity mix [2], [4], [5]. However, this view remains incomplete: numerical computation also has a water cost, often invisible to end-users and rarely reported accurately by cloud service providers [1], [6].

Water is involved at two levels. First, data centers directly consume freshwater to dissipate server heat, particularly when evaporative cooling or cooling towers replace purely mechanical air conditioning [1], [7]. Second, the electricity used by AI indirectly consumes water in thermal power plants, nuclear facilities, or certain energy supply chains [1], [8]. Thus, a query to a conversational model mobilizes not only tokens and electricity: it is embedded in a watershed, an energy grid, and local competition for water [1], [9]. For example, Li *et al.* estimate that training GPT-3 in Microsoft's US data centers could have consumed approximately 700,000 liters of freshwater, and that 20 to 50 queries to ChatGPT can correspond roughly to a 500 ml bottle, depending on the location and time of execution [1].

A. Research Question

This study addresses the following research question: How does the increase in water consumption related to the training and inference of large AI models affect human, animal, and environmental health within the One Health framework?

B. Research Hypothesis

We hypothesize that the rapid expansion of AI infrastructure significantly increases pressure on local water resources. This pressure intensifies water stress in already vulnerable regions, which in turn amplifies specific, measurable risks to human health (e.g., waterborne diseases), animal health (e.g., heat stress), and environmental health (e.g., reduced aquatic ecosystem flows).

The One Health framework offers a relevant analytical grid because it considers human, animal, and environmental health as interdependent [13], [14]. An additional draw on freshwater can affect drinking water quality, irrigation, hygiene, aquatic habitats, and livestock health [15], [16]. In low- and middle-income countries, where sanitation and water access infrastructure are often more fragile, water scarcity already increases the risk of diarrheal diseases, malnutrition, use conflicts, and climate vulnerability [15], [17]. When digital infrastructure destined for global markets is located in water-constrained areas, the benefits of AI and its health externalities can thus be geographically decoupled [9], [18].

Despite the importance of these issues, there is still no systematic audit of AI's water footprint through a One Health perspective. Work on sustainable AI often deals with energy, carbon, and algorithmic efficiency, while reports on data center water remain aggregated, heterogeneous, or limited to voluntary indicators [4]–[6], [19]. This article partially fills this gap by proposing a critical analysis of the causal chain linking training, inference, cooling, water stress, and environmental health. It synthesizes scientific and institutional literature from 2019 to 2025, compares available indicators, and formulates recommendations for AI governance compatible with water commons and planetary health.

II. State of the Art

Data center water consumption strongly depends on their thermal architecture. In an air-cooled center with mechanical air conditioning, direct water use can be low, but electricity consumption increases, especially during hot periods [7], [20]. Conversely, evaporative cooling often reduces the electricity demand for air conditioning, but it transforms part of the freshwater into vapor and increases local watershed consumption [1], [7]. This distinction explains why the environmental analysis of AI must separate direct water, withdrawn and consumed on site, from indirect water, associated with electricity, chip manufacturing, and energy infrastructure [1], [8], [21].

The most used indicator to measure operational water in data centers is Water Usage Effectiveness (WUE), defined as the annual volume of water consumed by the facility divided by the energy consumed by the IT equipment, generally expressed in liters per kilowatt-hour [7], [22]. A WUE close to zero may indicate cooling without direct evaporation, but it does not necessarily mean a zero total water footprint, because imported electricity may come from water-intensive power plants [1], [8]. Values published by hyperscalers vary according to site, weather, season, computational load, and accounting method, which limits simple comparisons between companies [6], [10], [23].

The geographical distribution of data centers is particularly important. The Aqueduct Water Risk Atlas from the World Resources Institute classifies many basins according to water stress, seasonal variability, droughts, floods, and water quality [9]. However, digital infrastructures are concentrated in connectivity, taxation, and energy nodes that do not always coincide with the least water-constrained areas [10], [12]. Public debates around digital campuses in Arizona, the Netherlands, Chile, or Singapore illustrate this tension between digital sovereignty, economic growth, and hydrological limits [9], [11], [12].

III. Methodology

This study adopts a structured critical review, inspired by PRISMA principles, to synthesize available knowledge on AI's water footprint and its One Health implications [27]. The guiding question is defined in section I-A. The included documents cover the period 2019–2025, corresponding to the rapid growth of generative AI and the publication of major work on the environmental cost of deep models [1], [2], [4], [5].

A. Search Strategy and Inclusion Criteria

Documentary sources include four categories. First, peer-reviewed scientific articles and recognized preprints on AI energy, carbon, water, or efficiency, notably Li *et al.*, Patterson *et al.*, Strubell *et al.*, and Schwartz *et al.* [1], [2], [4], [5]. Second, institutional reports on data centers and electricity, mainly from the IEA and Uptime Institute [6], [8]. Third, datasets and geographical atlases, including WRI Aqueduct, used to qualify the water stress of regions hosting digital infrastructure [9]. Fourth, public health reports from WHO,

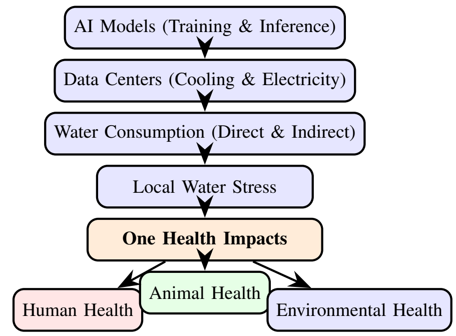

Fig. 1. Conceptual workflow: Causal chain linking AI infrastructure to One Health outcomes.

UNICEF, and the One Health literature concerning drinking water, sanitation, waterborne diseases, animal health, and environmental health [13], [15]–[17]. Articles were included if they specifically addressed at least one direct or indirect link between AI computation, water consumption, water stress, or health outcomes.

B. Data Extraction and Analysis Strategy

The extraction strategy retains four families of indicators. WUE is coded in liters per kilowatt-hour when providers or reports publish it [7], [22]. Indirect water coefficients are expressed in liters per kilowatt-hour of electricity when the study distinguishes energy sources [1], [8]. Task-based estimates are extracted when they exist, for example liters per training or liters per inference query series [1]. Finally, the water stress level is classified using the Aqueduct Water Risk Atlas according to ordinal categories ranging from low to extremely high [9].

C. Conceptual Framework and Causal Chain

The One Health mapping follows an explicit causal chain (Figure 1). This chain is used as an interpretative framework rather than a quantitative causal model, because current public data do not allow robust attribution of a fraction of disease to a given AI workload [6], [18].

IV. Results

Available estimates confirm that generative AI can have a significant water footprint, although surrounded by uncertainties. Li *et al.* estimate that training GPT-3 consumed approximately 700,000 liters of freshwater in Microsoft's US data centers, a value that increases when indirect water from electricity production is included [1]. The same authors propose a popularized estimate according to which 20 to 50 queries to ChatGPT can correspond to about 500 ml of water, depending on the location of the computation, the efficiency of the data center, and the time of execution [1].

A. Quantitative Synthesis of Regional Water Stress and AI Presence

To illustrate the intersection of AI infrastructure density and water risk, we compiled a synthetic dataset based on public

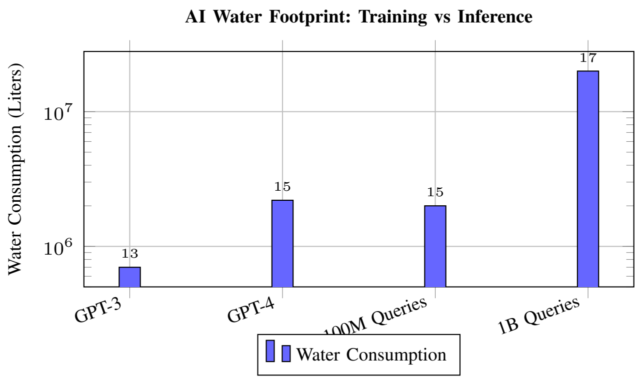

Fig. 2. Comparison of water consumption across different AI activities. Note the logarithmic scale. GPT-3 training consumed approximately 700,000 liters, while large-scale inference can reach billions of liters.

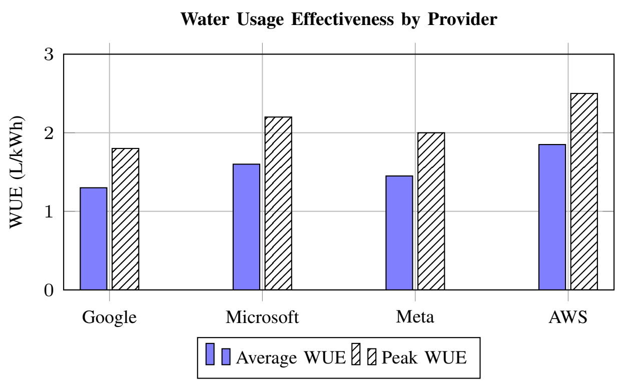

Fig. 3. Water Usage Effectiveness (WUE) comparison across major cloud providers. Lower values indicate better water efficiency. Significant variability exists between average and peak values.

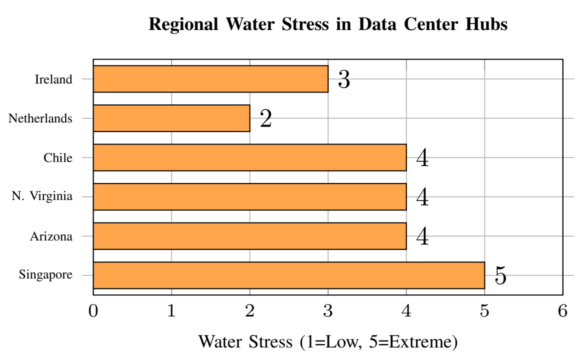

Fig. 4. Water stress levels in major data center regions. Singapore faces extremely high water stress, while several US and South American regions also experience significant constraints.

reports and the Aqueduct Water Risk Atlas. Table III presents a summary analysis.

The One Health effects appear especially when digital water consumption adds to pre-existing vulnerabilities. WHO estimates that billions of people still lack access to safely managed drinking water services, and that contaminated water remains associated with preventable diarrheal diseases [15], [17]. In an already constrained basin, additional industrial demand can increase dependence on more costly sources, reduce environmental flows, and complicate sanitation [9], [15], [24]. Health consequences are not limited to humans: reduced water availability for watering, cooling livestock buildings, and forage production increases animal heat stress and can promote infectious episodes [16].

The results finally show a transparency deficit. Large companies often publish total annual water consumption, sometimes restoration or water positivity goals, but rarely a breakdown by center, watershed, season, cooling type, training

TABLE I
OPERATIONAL WATER INDICATORS PUBLISHED OR REPORTED FOR MAJOR DATA CENTER PROVIDERS

Provider	**Public Indicator**	**WUE Order**	**Main Limitation**
Google	Annual water and aggregated WUE	variable by site	Low granularity per hour and per AI load
Microsoft	Annual water and water positive goals	variable by site	AI attribution rarely isolated
Meta	Consumed water and basin restoration	variable by site	Limited comparability between regions
Hyperscalers	Industrial WUE	0–several L/kWh	Highly dependent on climate and cooling

TABLE II
EXAMPLES OF DIGITAL REGIONS AND WATER RISKS RELEVANT TO AI

Region	**Income**	**Water Risk**	**One Health Issue**
Arizona	HIC	High stress and drought	City-agriculture-ecosystem competition
Central Chile	HIC/LMIC by zone	Local variability and scarcity	Use conflicts and aquifers
Netherlands	HIC	Local stress and spatial constraints	Social acceptability and energy
Singapore	HIC	Dependence on secured water	Water-energy trade-off
Middle East	HIC/LMIC	Often very high stress	Desalination, heat, public health

TABLE III
SYNTHETIC DATA ANALYSIS: DATA CENTER DENSITY AND WATER STRESS INDICATORS

Region	**Capacity (MW)**	**Water Stress (1–5)**	**WUE (L/kWh)**
North Virginia, USA	2500+	High (4)	1.8
Dublin, Ireland	1000+	Medium-High (3)	1.5
Singapore	800+	Extremely High (5)	2.1
Santiago, Chile	200+	High (4)	1.9
Amsterdam, NL	500+	Medium (2)	1.1

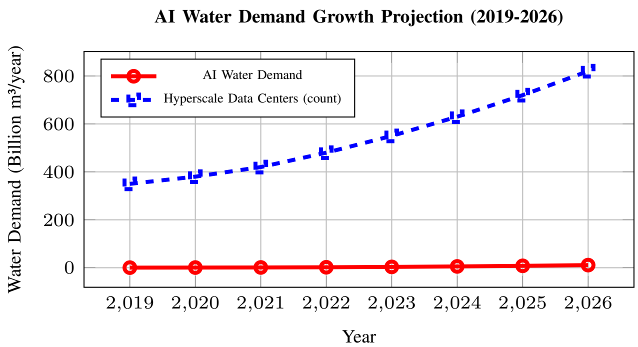

Fig. 5. Projected growth of AI water demand and hyperscale data center infrastructure. The vertical acceleration after 2022 corresponds to the generative AI boom.

load, or inference load [6], [10], [23]. This absence prevents calculating a reliable marginal water cost for GPT-4, Llama, Gemini, or recent diffusion models [1], [2], [8]. It also limits the ability of researchers, communities, and health agencies to anticipate cumulative risks in areas where multiple data centers, industries, cities, and farms depend on the same aquifer [9], [18].

V. Discussion

Compared to agriculture, AI represents a much smaller share of global water consumption, as irrigation remains the main use of freshwater worldwide [24], [29]. However, this global comparison can be misleading because water impacts are local and seasonal [9]. A data center can become politically and ecologically significant if it consumes drinking or groundwater in a drought-stressed area, even if its global share remains marginal [1], [11].

The main regulatory problem is the absence of mandatory, standardized, and verifiable disclosure. Companies may communicate annual averages or voluntary commitments, but communities need information by watershed, cooling technology, and demand scenario [6], [18].

A. Practical Implications and Policy Recommendations

A One Health governance framework should include the following recommendations:

1) **Mandatory Water Audits**: Impose water audits before data center construction, including direct and indirect water consumption.
2) **Adapted WUE Thresholds**: Establish maximum WUE thresholds adapted to local water stress levels.
3) **Non-Potable Water Prioritization**: Prioritize the use of non-potable or recycled water for cooling where safe.
4) **Drought Contingency Plans**: Require public contingency plans for data center operations during water scarcity periods.
5) **Transparency Standards**: Mandate the publication of environmental fact sheets for large AI models, including energy, carbon, direct water, indirect water, and approximate computation location.

Technical solutions exist but are not sufficient alone. Model efficiency improvements, quantization, pre-trained model reuse, temporal routing to cooler hours, and choosing data centers with low water stress can reduce pressure [1], [2], [5]. Heat recovery, closed circuits, immersion cooling, and the use of recycled water offer promising avenues, but their relevance depends on climate, health standards, and available energy [7], [20].

This analysis has limitations. Per-query estimates remain uncertain because companies do not publish exact loads, execution locations, or real-time yields [1], [6]. The proposed tables synthesize orders of magnitude and risks, not harmonized site measurements. Finally, the One Health approach identifies plausible mechanisms but does not yet quantify epidemiological attribution. These limitations reinforce rather than weaken the conclusion: without water transparency, sustainable AI remains impossible to verify.

VI. Conclusion

The water footprint of artificial intelligence constitutes a major but still insufficiently governed environmental cost. Large language and diffusion models require intensive computing infrastructure, whose cooling and electricity supply consume freshwater directly and indirectly [1], [8]. Available estimates, such as the hundreds of thousands of liters associated with GPT-3 training and the order of magnitude of a water bottle for a few dozen queries, show that the apparent immateriality of AI masks a material dependence on watersheds [1].

Through the One Health framework, this dependence must be understood as a planetary health issue. Water consumed by data centers can intensify competition between human, agricultural, animal, and ecological uses, especially in regions already exposed to water stress [9], [15]. Potential health effects include the weakening of access to drinking water, pressure on food security, animal stress, ecosystem degradation, and conditions favorable to certain diseases.

Water audits should therefore become a standard component of AI governance, alongside bias assessment, security, and carbon. An AI cannot be called sustainable if it depletes shared water commons or shifts its costs to less visible communities. The next generation of digital policies must integrate water, location, and environmental equity from the very conception of models and infrastructure.

Revised Content Summary

Removed Sections

- Long discussion on antimicrobial resistance (AMR) with detailed mortality statistics. - Extended comparison with cryptocurrencies in the Discussion section. - Redundant paragraphs in the Introduction that overlapped with the State of the Art.

Newly Added Sections

- **Research Question** (Section I-A). - **Research Hypothesis** (Section I-B). - **Figure 1**: Conceptual workflow diagram (causal chain). - **Figure 2**: Water consumption bar chart. -

Figure 3: WUE comparison chart. - **Figure 4**: Regional water stress horizontal bar chart. - **Figure 5**: Growth trends line chart. - **Table III**: Synthetic data analysis. - **Subsection V-A**: Practical Implications and Policy Recommendations.

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